SAY WATT ROBOTICS · FIELD LOG

Week 1.13 — Why a Company Has a Name — Liability, a Gripper That Closes, and a Shopping List for the CFO

Sunday, September 13, 2026 · 3:00 – 5:00 PM · 13 students

Thirteen of us at 1.13 — Trenton, Isaac, Liam, Annabelle, Lillian, Tyler, Evan, Oliver, Eli, Frankie, Sam, Rebecca, and Rhylan — from 3:00 to 5:00. Every track moved, but the moment worth writing down came at the closeout, when the company name came back around and the question underneath it got asked out loud: why does a company exist in the first place? The answer is liability. Build something under your own name and your own Social Security number, and if somebody is hurt using it, they sue you — your house, your savings, everything you have already earned. Build it under a company, and they sue the company. The company can fail. You do not. That is what the letters in LLC actually mean, and it is why MATE teams compete as companies rather than as clubs: the name on the technical documentation is a real thing, not decoration. Two candidates are on the board — Aquassential Systems and NewClear Systems — and the decision now has a date on it: September 30. It came with a second lesson, which is how to triage a decision. Is it time-sensitive? Not today. Is it important? Yes. That combination is exactly the kind that quietly slips, so it gets a deadline instead of a shrug. On the build side, the payload track has something real: a hand gripper driven by a linear actuator, prototyped and finished this session, with a control program written to drive it. The hydraulic concept gets built too, and then both get demonstrated side by side so the trade study is settled by watching rather than arguing. The sensor track closed its recommendation and handed it to the CFO as a proposal he can act on — a pressure sensor, an inertial measurement unit, and a camera. Management built a prototype budget prioritization so the money goes to whatever has to be proven first, and pushed the integrated master schedule toward final. Electrical kept the schematic moving in KiCad. And marketing has the STEAM night slide show for Rainbow Elementary essentially done, with fresh photos in it.

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STUDENTS AT THE SESSION	SENSORS ON THE PROPOSAL TO THE CFO	CANDIDATE COMPANY NAMES ON THE BOARD	DAYS TO DECIDE THE NAME

WHAT WE LEARNED

- Why businesses incorporate, in plain terms: build something under your own name and a person hurt using it sues you personally — your house, your savings, everything already earned. Build it under a company and they sue the company. The company can fail; you do not. That is what limited liability means.
- That is also why MATE teams compete as companies instead of clubs. The name goes on the technical documentation, the marketing display, and the poster at the regional — it is the entity doing the work, not a label on it.
- Two questions triage any decision: is it time-sensitive, and is it important? Important-but-not-urgent is the dangerous box, because nothing forces it and it slips. The fix is to put a date on it — which is how the company name got September 30.
- A prototype ends an argument a whiteboard cannot. Build the linear-actuator gripper, build the hydraulic one, run both, and let the trade study report what happened rather than what everyone expected.
- Prioritize the prototype money before you spend it. Not everything needs proving, and the things that do should be bought in the order the risk demands.
- A parts list becomes a purchase when somebody owns the decision. The sensor track's real job was not picking sensors — it was handing the CFO a proposal complete enough to act on.

DECISIONS MADE

- The sensor recommendation is closed and goes to the CFO as a proposal, not a wish: a pressure sensor, an inertial measurement unit, and a camera. It carries the head-to-head comparison work from 1.11 and 1.12 behind it, which is what makes it a proposal somebody can sign.
- The manipulator gets decided by demonstration. The linear-actuator hand gripper is prototyped and has code driving it; the hydraulic concept gets built next, and both get shown side by side before the trade is written down.
- Prototype money gets prioritized before anything is ordered. Management built a ranked list of what has to be proven first, so the budget buys certainty in the order the team needs it.
- The integrated master schedule stays a management priority and moved toward final this session. (It was finished and published at 2.7, on November 1.)
- The company name is a scheduled decision, not an open one. Two candidates, a September 30 deadline, and the reasoning for why the name matters now on the record.

ACTION ITEMS

- Finalize the company name and slogan
  Trenton (CEO) · Sept 30
- Buy the sensors and the prototyping parts against the prioritized budget
  Maddox (CFO) · by 1.14
- Order the sensors so they can be tested during the prototype phase
  Rhylan (Sensor) · by 1.14
- Build the hydraulic manipulator, demonstrate it against the linear actuator, write the trade
  Eli (Payload) · by 1.14
- Finish the electrical schematic
  Liam (Electrical) · by 1.14
- Keep the Arduino integration research going
  Oliver (Software) · by 1.14
- Finalize the motor control type
  Oliver (Software) · by 1.14
- Stand up the team Instagram account (carried from 1.12)
  Lillian (Marketing) · by 1.14
- RSVP yes or no on GroupMe — no maybes
  Every student · this week

STUDENT LEARNING OUTCOMES

- Learned what an LLC is and why engineers and inventors incorporate — limited liability explained in terms a builder can use.
- Closed a sensor proposal — pressure sensor, inertial measurement unit, and camera — and handed it to the CFO as a purchase recommendation.
- Built and finished a payload manipulator prototype: a hand gripper driven by a linear actuator.
- Wrote the control program that drives the manipulator prototype.
- Continued the TriggerFish control-box schematic in KiCad.
- Built a prototype budget prioritization — deciding which components get proven first, and with what money.
- Worked the integrated master schedule toward final.
- Finished the STEAM night slide show for Rainbow Elementary and refreshed the photographs in it.
- Brought two candidate company names to the team and learned how to put a deadline on an important decision.

LOOKING AHEAD

The name has a date now — September 30, two candidates on the board. Of the four items carried into this session, the integrated master schedule is nearly there and the name is scheduled; capsule dive #3 (the gate is three consecutive dry dives) and the first System Integration Diagram walkthrough are still open, along with the Phase 1 showcase and role retrospective. Near-term: get the sensors ordered so the prototype phase can actually start, build the hydraulic gripper and run it against the linear actuator with the trade written down, finish the electrical schematic, and settle the motor control type. And everybody RSVP on GroupMe — yes or no, no maybes.

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